

Quiz

May. 25, 2026

多选题 2分

Which of the following are **components or responsibilities** of a file system?

- ☐ A Naming (interface to find files by name)
- ☐ B Disk management (collecting disk blocks into files)
- ☐ C Process scheduling
- ☐ D Protection (keeping data secure)
- ☐ E Reliability/durability (keeping files durable despite crashes)

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多选题 2分

Which of the following are **problems or characteristics of contiguous allocation**?

- ☐ A External fragmentation
- ☐ B Poor random access performance
- ☐ C Defragmentation can help but is expensive
- ☐ D Internal fragmentation
- ☐ E Simple file creation and deletion

提交

Which statements about the **FAT (File Allocation Table) series** are correct?

- A** FAT uses a linked-list-like structure to track file blocks
- B** FAT32 supports files up to 4GB - 1 byte
- C** Deleting a file in FAT only marks the directory entry as free (0xE5)
- D** FAT provides built-in encryption and access control
- E** FAT is commonly used in SD cards and USB drives

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多选题 2分

Which of the following are **true about Unix/Linux inodes (Ext2/Ext3)**?

- A** An inode contains file metadata (owner, size, permissions, etc.)
- B** Small files use only direct block pointers
- C** Large files use indirect, double-indirect, or triple-indirect pointers
- D** The inode table size is fixed, limiting the maximum number of files
- E** Hard links create a new inode

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多选题 2分

Which of the following correctly describe **hard links vs. symbolic (soft) links**?

- A** Hard links cannot cross filesystem boundaries
- B** Symbolic links remain valid if the target is moved
- C** Deleting a symbolic link leaves the target unchanged
- D** Hard links increment the link count in the inode
- E** Symbolic links can point to directories or files

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